เลขที่นั่งสอบ

College of Industrial Technology

King Mongkut's University of Technology North Bangkok

Final Examination of Semester 2 **Year:** 2013

Subject: 394172 Mathematics II **Section:** 15-16

Date: 5 March 2014 **Time:** 9.00-12.00

Name:______________________________**ID:**______________________**Class:**________

Instruction

1. Cheating will result in failure of all classes registered of the current semester. Students who are caught cheating will also be denied registering for the following semester.
2. No documents are allowed to be taken out of the examination room.
3. Textbooks are **NOT** allowed.
4. Calculation devices are allowed.
5. Electronic communication devices are **NOT** allowed in the examination room.
6. The examination has 8 pages (including this page), with 12 questions and a total score of 120 marks.
7. Write all your answers on this exam sheets.

Question 1. Solving the system of equations. (10 marks)

$$\frac{7}{x}+\frac{5}{y}-\frac{7}{z}=-8$$

$$\frac{4}{x}+\frac{2}{y}-\frac{3}{z}=0$$

$$\frac{5}{x}-\frac{4}{y}+\frac{4}{z}=35$$

Question 2. Solving the absolutely inequality $\dfrac{|2-x|}{|1+x|-1} \leq 0$. (10 marks)

Question 3. To simplify the expression

$$\frac{1}{\sqrt{11-2\sqrt{30}}}-\frac{1}{\sqrt{7-2\sqrt{10}}}-\frac{1}{\sqrt{8+4\sqrt{3}}}$$ (10 marks)

Question 4. To Evaluate the radical expression $\sqrt{\sqrt{243}-\sqrt{240}}$ (10 marks)

Question 5. To solve the radical equation $\sqrt{3x+9} - \sqrt{x+5} = \sqrt{2x+8}$. (10 marks)

Question 6. Solve the exponential equality $\left(\frac{4}{5}\right)^x + \left(\frac{5}{4}\right)^x - 2.05 = 0$. (10 marks)

Question 7. Solve the exponential inequality $\left(\frac{1}{7}\right)^{x^2+2x+8} < 7^{-2x-24}$. (10 marks)

Question 8. Given $A = \log_a \frac{6}{5} - \log_a 300 + \log_a 125$ and

$B = \log_a \frac{9}{32} + 2\log_a \frac{256}{3} + \log_a \frac{3}{8} + \log_a \frac{1}{3}$. Evaluate the expression $\frac{A}{B}$. (10 marks)

Question 9. Find the solutions of the inequality

$$\log_{\frac{1}{4}} \log_{\frac{1}{3}} \log_{\frac{1}{2}} \sqrt[3]{\frac{1}{x^2-3x+4}} \ge 0$$ (10 marks)

Question 10. Find all integer solutions of the equation

$$9^{\log_{\frac{1}{3}}(x+2)} - 5^{\log_{\frac{1}{5}}(2x^2+1)} = 0$$ (10 marks)

Question 11. To solve the logarithmic equation

$$\log_4\left(2\log_3\left(1+\log_2(1+3\log_3 x)\right)\right) = \frac{1}{2}$$

(10 marks)

Question 12. Given the system of equations

$$\begin{aligned} 2^{2y-3} &= 3^{y-1} \\ 2(\log_2 0.7)(\log_{0.7} x) &= \log_2 4 \end{aligned}$$

What is the value of $|x+y|$?

(10 marks)